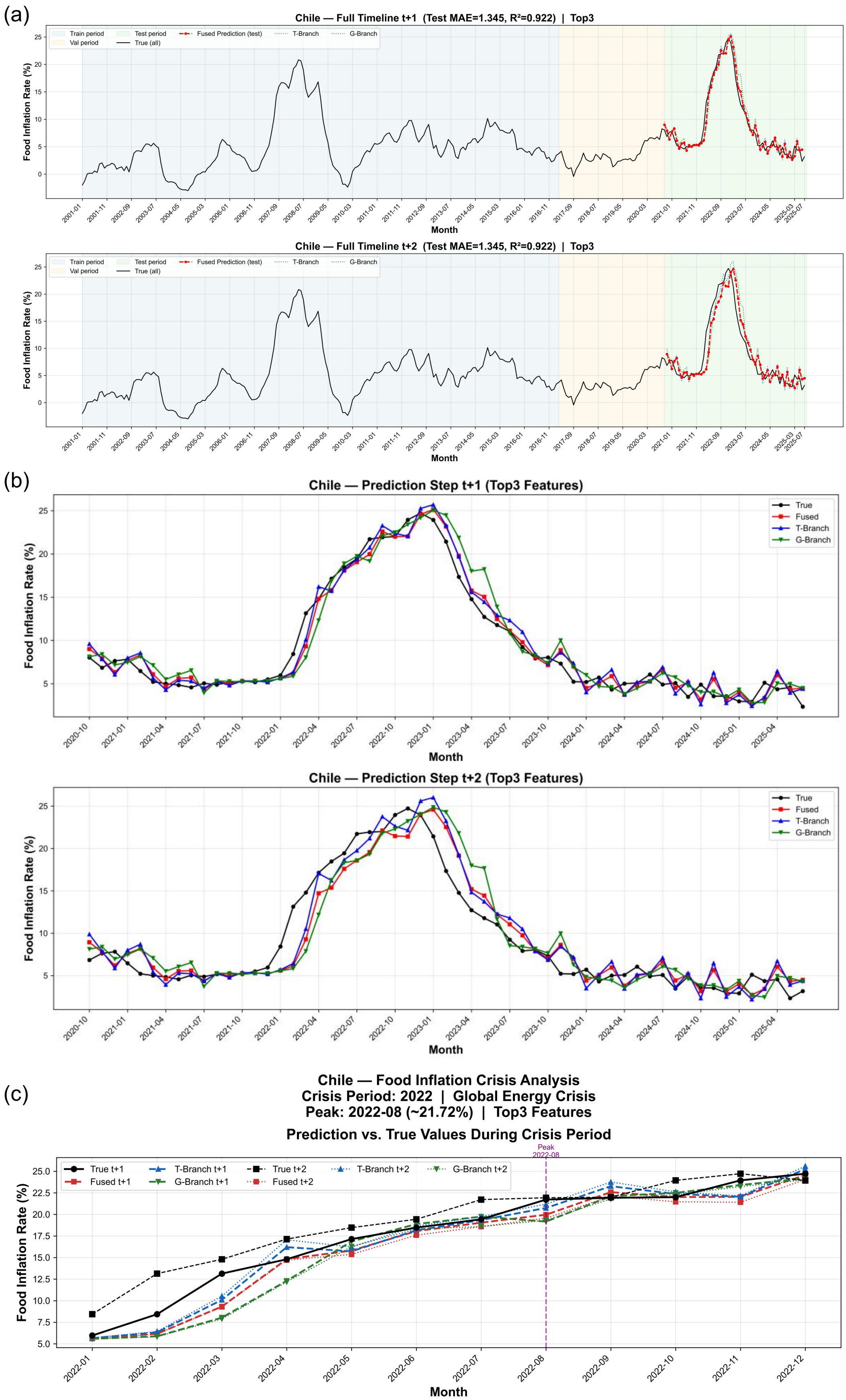
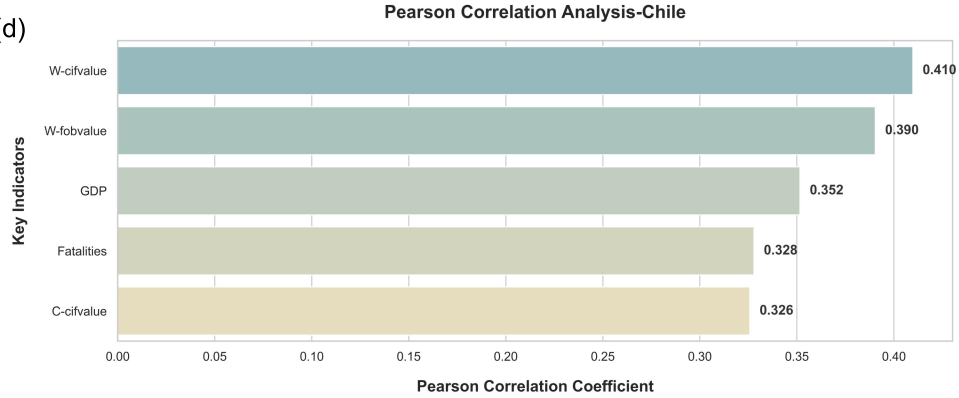
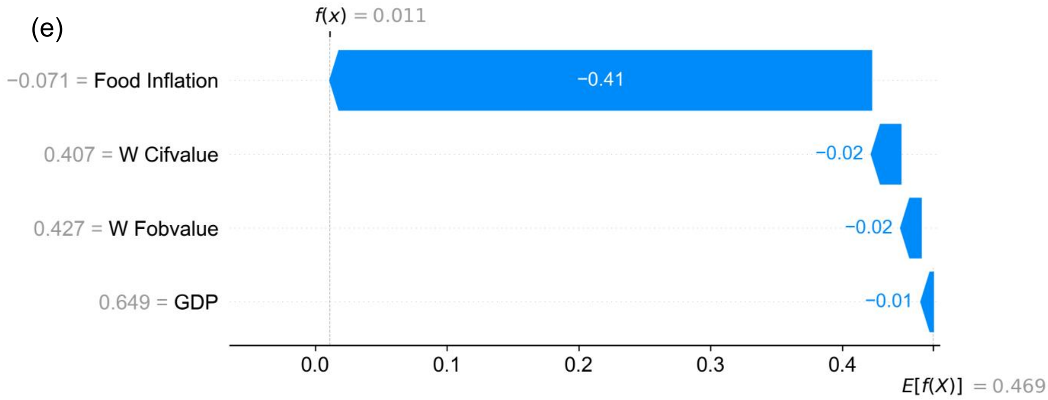

(a)
Chile — Full Timeline t+1 (Test MAE=1.345, R²=0.922) | Top3
Train period
Val period
Test period
True (all)
Fused Prediction (test)
T-Branch
G-Branch
Food Inflation Rate (%)
Month
Chile — Full Timeline t+2 (Test MAE=1.345, R²=0.922) | Top3
(b)
Chile — Prediction Step t+1 (Top3 Features)
True
Fused
T-Branch
G-Branch
Chile — Prediction Step t+2 (Top3 Features)
(c)
Chile — Food Inflation Crisis Analysis
Crisis Period: 2022 | Global Energy Crisis
Peak: 2022-08 (~21.72%) | Top3 Features
Prediction vs. True Values During Crisis Period
Peak
2022-08
True t+1
Fused t+1
T-Branch t+1
G-Branch t+1
True t+2
Fused t+2
T-Branch t+2
G-Branch t+2

(d)



(e)



Multidimensional Evaluation and Attribution Analysis of Food Inflation Forecast in Chile. (a) Overview of the total cycle prediction and fitting, (b) Multi-step Short-Term Forecast Comparison Chart, (c) Details of Prediction Performance in Extreme Crisis Periods, (d) Pearson correlation analysis of macroeconomic drivers, (e) SHAP-based explanation of feature contributions for a specific prediction instance.

The direct transmission of exchange rate fluctuations and the output gap driven by prolonged droughts form the primary axis of food inflation in Chile. As a representative export-oriented economy, the Chilean peso's exchange rate against the U.S. dollar directly dictates the cost of imported meat, dairy, and grains, where currency weakening rapidly triggers imported inflation. On the production side, Chile's central agricultural belt is enduring a decadal drought; water scarcity has inflated irrigation costs and curtailed fruit and vegetable yields, creating a double squeeze between the shrinking domestic supply and soaring import costs. Furthermore, Chile’s highly market-based pricing system ensures that international energy price hikes are instantly reflected in domestic logistics surcharges, increasing distribution costs across its narrow territory. This combination of high external dependence and local ecological fragility renders Chile's food prices exceptionally sensitive to macroeconomic volatility.